

- (d) Ring counter
- (e) Twisted ring counter
- (f) Sequence counter

4.11. Asynchronous Counter Or Ripple Counter

A circuit which is used for counting the numbers or pulses is known as counter. Counter is referred to as modulo-N (or divide by N), where the word modulo indicates the number of states in the counter.

4.11.1. 3- Bit Binary Counter

Consider a 3-bit binary counter which has total '8' number of states which require three flip-flops and Q_2 , Q_1 and Q_0 are the outputs of those flip-flops.

The circuit diagram or logic circuit diagram for 3-bit binary counter,

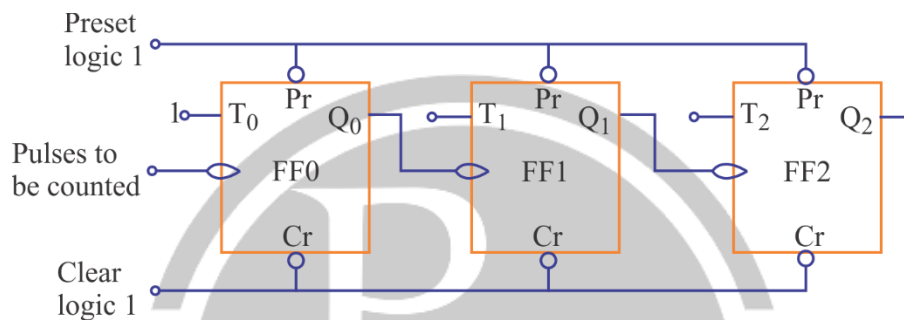


Fig. 4.31. A 3-bit Binary Counter

The truth table for 3-bit binary counter is given as:

Counter state	Count		
	Q_2	Q_1	Q_0
0	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0	1
6	1	1	0
7	1	1	1

Output waveforms of the above counter is:

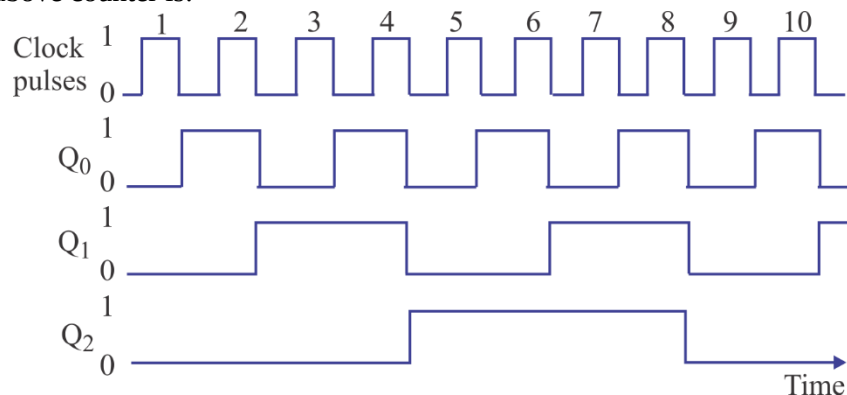


Fig. 4.32.

The frequency ' f ' of clock pulses for reliable operation of the counter is given as

Where, N = number of flip-flops

t_{pd} = propagation delay of one flip-flop.

T_s = strobe pulse width.

If during the operation of counter, if some pulses are falsely operated for short duration, known as spikes, which change the state of the flip-flop. It may happen when the propagation delay of each flip-flop may vary and may happen that, all the flip-flops may not change their states or may be only one flip-flop changes its state during the pulse time.

This problem of spikes can be eliminated by using a strobe pulse with the help of strobe pulse, the state will change only when flip-flops of the counter are in steady state.

Example: In a 4-stage ripple counter, the propagation delay of a flip-flop is 50n sec. If the pulse width of the strobe is 30n sec. Find the maximum frequency at which the counter operator reliably.

Solution: The maximum frequency is

$$f_{\max} = \frac{1}{nt_{pd} + t_s}$$

n = number of flip-flops or stage = 4

t_{pd} = propagation delay of each flip-flop = 50 nsec.

t_s = Strobe pulse width = 30 nsec

$$f_{\max} = \frac{1}{(4 \times 50 + 30) \times 10^{-9}} = \frac{1000}{(200 + 30)} \text{ MHz} = \frac{1000}{230} \text{ MHz}$$

4.11.2. Modulo-6 Asynchronous Down Counter

Down counter is the counter which counters the values of pulses in descending order. Consider or Stable-8 counter ($2^3 = 8$), which uses three flip-flops.

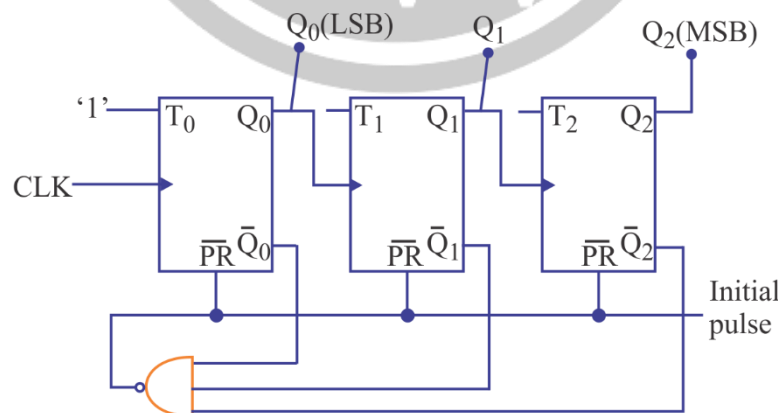


Fig. 4.33.

1. Only sequential counter can be designed. Random counter cannot be designed.
2. Glitch (undesirable state) would appear in case of asynchronous counter.
3. Speed of asynchronous counter is not fast.

4.11.5. BCD Ripple Counter

A decimal counter follows a sequence of 10 states and returns to 0 after the count of 9. Such a counter must have at least four flip-flops to represent each decimal digit, since a decimal digit is represented by a binary code with at least four bits. The sequence of states in a decimal counter is dictated by the binary code used to represent a decimal digit. If the BCD code is used, the sequence of states is as shown in the state diagram. A decimal counter is similar to a binary counter, except that the state after 1001 (the code for decimal digit 9) is 0000 (the code for decimal digit 0).

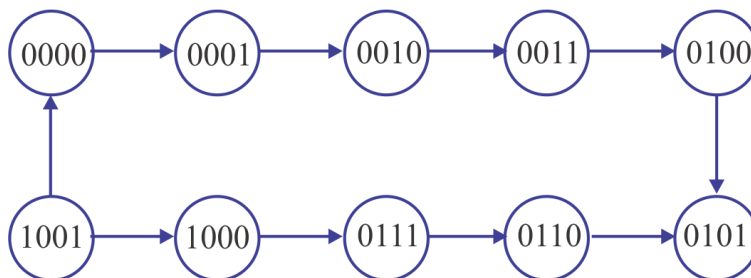


Fig. 4.34. State diagram of a decimal BCD counter.

The logic diagram of a BCD ripple counter using JK flip-flops is shown in figure below. The four outputs are designated by the letter symbol Q, with a numeric subscript equal to the binary weight of the corresponding bit in the BCD code. Note that the output of Q₁ is applied to the C inputs of both Q₂ and Q₄ and the output of Q₂ is applied to the C and output of Q₂ and Q₃ applied to J through a two input AND gate.

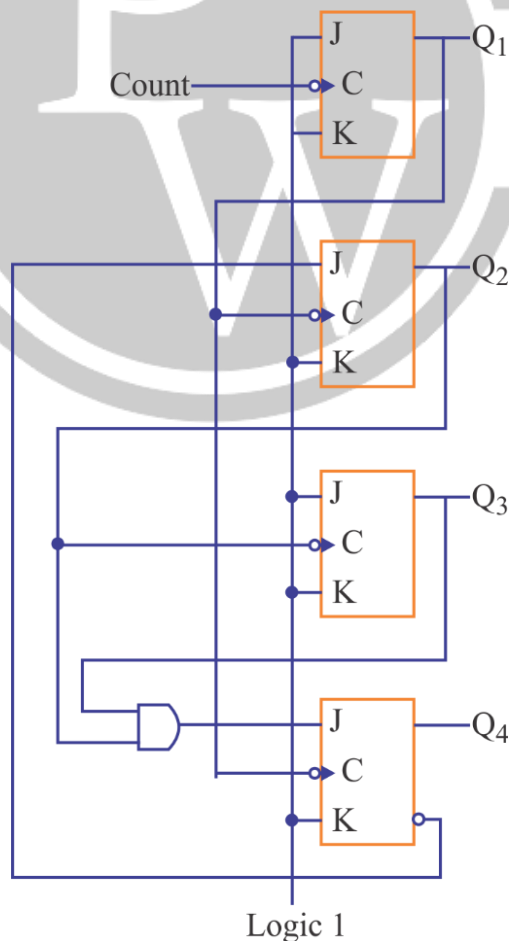


Fig. 4.35. BCD ripple counter

4.12. Synchronous Counter

The ripple counters have the advantage of simplicity (only FLIP-FLOP's are required) but their speed is low because of ripple action. The maximum time is required when the output changes from 111....1 to 00....0 and this limits the frequency of operation of ripple counters.

The speed of operation improves significantly if all the FLIP-FLOPS are clocked simultaneously. The resulting circuit is known as a synchronous counter. Synchronous counters can be designed for any count sequence (need not be straight binary).

The output Q_0 of the least-significant FLIP-FLOP changes for every clock pulse. This can be achieved by using a T-type FLIP-FLOP with $T_0 = 1$. The output Q_0 changes whenever Q_0 changes from 1 to 0. Therefore, if Q_0 is connected to T input (T_1) of the next FLIP-FLOP, Q_1 will change from 1 to 0 (or 0 to 1) when $Q_0 = 1$ ($T_1 = 1$) and will remain unaffected when $Q_0 = T_1 = 0$. Similarly, Q_2 changes whenever Q_1 and Q_0 are both "1". This can be achieved by making the T-input (T_2) of the most-significant FLIP-FLOP equal to $Q_1 \cdot Q_0$.

In addition to FF's, synchronous counters require some gates also. JK FLIP-FLOPS are the most commonly used FLIP-FLOP's for the design of synchronous counters. In this, each FLIP-FLOP has two control inputs (J and K) and circuit is required to be designed for each control input. Many programmable logic devices (PLDs) used for the design of digital systems utilise D FLIP-FLOPS for their memory elements, therefore, counter design using D FLIP-FLOPS will be useful for programming inside a PLD. It has only one control input which makes its design simpler than the design using J-K FLIP-FLOPS.

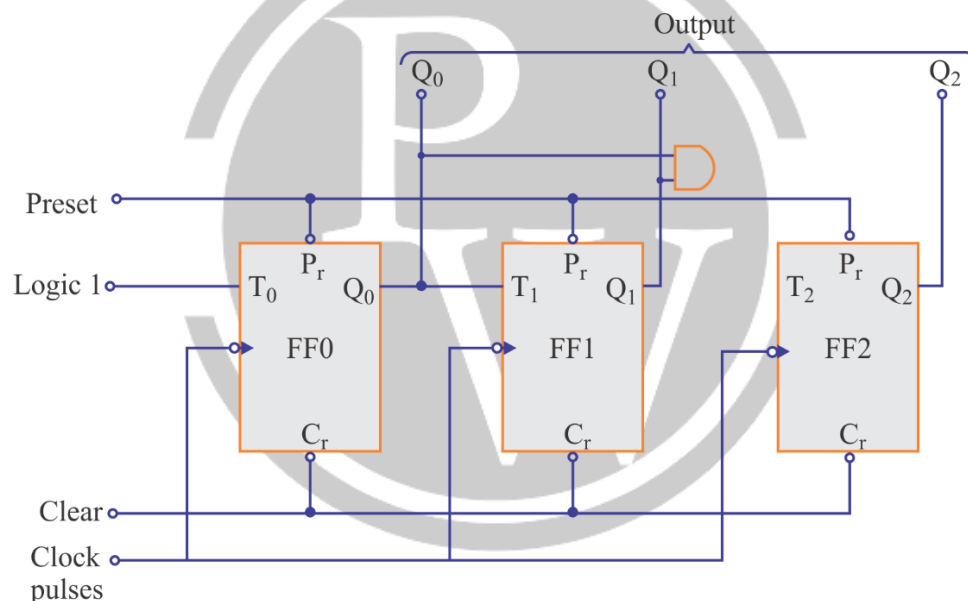


Fig. 4.36. A 3-bit Synchronous Counter

4.12.1. Synchronous Counter Design

Synchronous counters for any given count sequence and modulus can be designed in the following way:

1. Find the number of FLIP-FLOPs required.
2. Write the count sequence in the tabular form.
3. Determine the FLIP-FLOP inputs which must be present for the desired next state from the present state using the excitation table of the FLIP-FLOPS.
4. Prepare K-map for each FLIP-FLOP input in terms of FLIP-FLOP outputs as the input variables.
5. Simplify the K-maps and obtain the minimized expressions.
6. Connect the circuit using FLIP-FLOPS and other gates corresponding to the minimized expressions.

Example: Design a 3-bit synchronous counter using JK Flip-Flops.

Solution: The number of FLIP-FLOPs required is 3. Let the FLIP-FLOPs be FF0, FF1, FF2 and their inputs and outputs are given below:

FLIP-FLOP	Inputs	Outputs
FF0	J_0, K_0	Q_0
FF1	J_1, K_1	Q_1
FF2	J_2, K_2	Q_2

The count sequence and the required inputs of FLIP-FLOPs is shown below.

			FLIP-FLOP INPUTS					
Counter state			FF0		FF1		FF2	
Q_2	Q_1	Q_0	J_0	K_0	J_1	K_1	J_2	K_2
0	0	0	1	X	0	X	0	X
0	0	1	X	1	1	X	0	X
0	1	0	1	X	X	0	0	X
0	1	1	X	1	x	1	1	X
1	0	0	1	X	0	X	X	0
1	0	1	X	1	1	X	X	0
1	1	0	1	X	X	0	X	0
1	1	1	x	1	X	1	x	1
0	0	0						

$Q_2 Q_1$	00	01	11	10
Q_0				
0	1	1	1	1
1	x	x	x	x

$$J_0 = 1$$

(a)

$Q_2 Q_1$	00	01	11	10
Q_0				
0	x	x	x	x
1	1	1	1	1

$$K_0 = 1$$

(b)

$Q_2 Q_1$	00	01	11	10
Q_0				
0	0	x	x	0
1	1	x	x	1

$$J_1 = Q_0$$

(c)

$Q_2 Q_1$	00	01	11	10
Q_0				
0	x	0	0	x
1	x	1	1	x

$$K_1 = Q_0$$

(d)

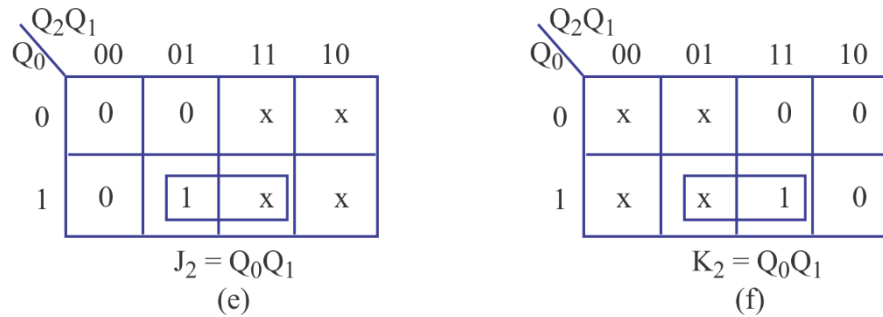


Fig. 4.37. K-Maps of 3-bit Synchronous Counter

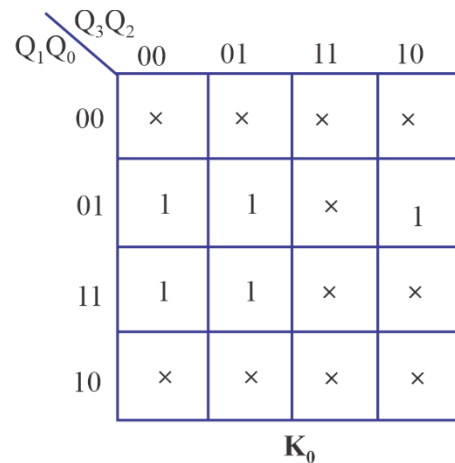
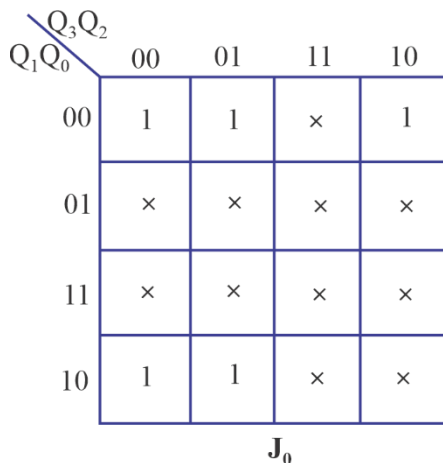
Example:

Design a natural binary sequence mod-8 synchronous counter using D FLIP—FLOPS.

Solution:

The number of FLIP-FLOPS required is 3. Let the FLIP—FLOPS be FF0, FF1 and FF2 with inputs D_0 , D_1 and D_2 , respectively. Their outputs are Q_0 , Q_1 , and Q_2 respectively

Counter State			FLIP-FLOP inputs		
Q_2	Q_1	Q_0	D_0	D_1	D_2
0	0	0	1	0	0
0	0	1	0	1	0
0	1	0	1	1	0
0	1	1	0	0	1
1	0	0	1	0	1
1	0	1	0	1	1
1	1	0	1	1	1
1	1	1	0	0	0



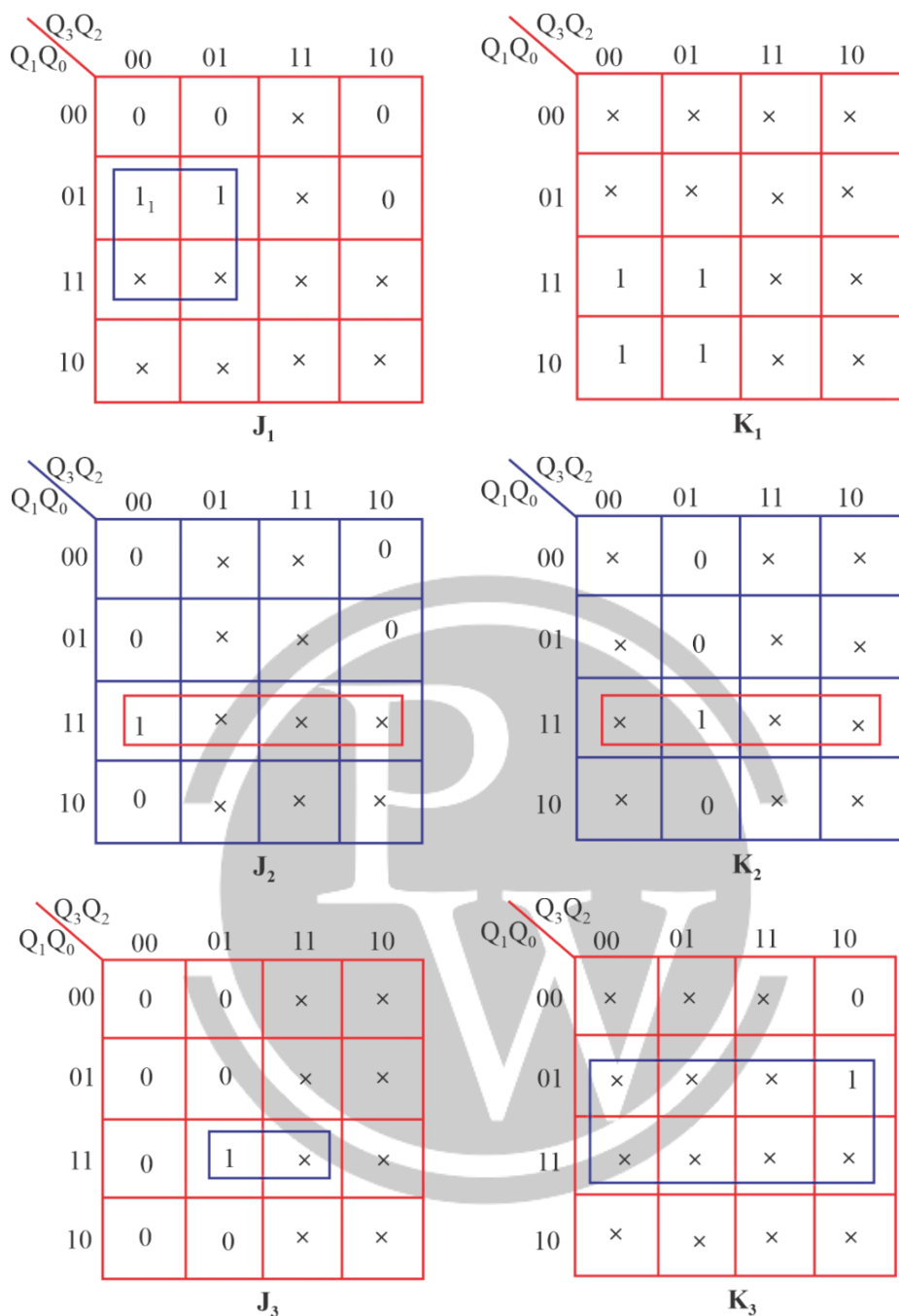


Fig. 4.38. K-Maps for 8-bit Synchronous counter

The K – maps for D_0 , D_1 and D_2 are given as,

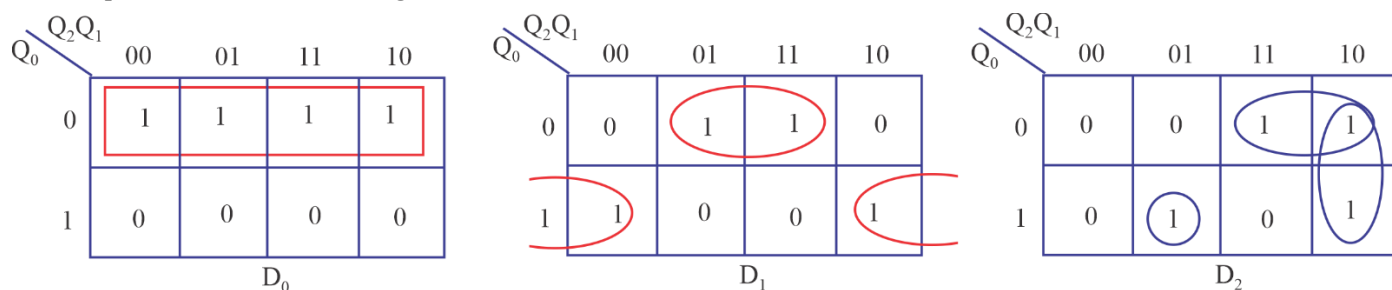


Fig. 4.39.